

INMOTC 2025 (MP region)

ALGEBRA

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§1 Problems

Exercise 1.1 (MMO 1946 Grades 7-8 P5). Prove that after completing the multiplication and collecting the terms

$$(1 - x + x^2 - x^3 + \cdots - x^{99} + x^{100})(1 + x + x^2 + \cdots + x^{99} + x^{100})$$

has no monomials of odd degree.

Summary — What happens if x is replaced by $-x$?

Walkthrough —

(a)

Solution 1. Let $P(x)$ denote the polynomial

$$(1 - x + x^2 - x^3 + \cdots - x^{99} + x^{100})(1 + x + x^2 + \cdots + x^{99} + x^{100}).$$

Note that $P(x) = P(-x)$. By the Claim below, it follows that $P(x)$ has no monomials of odd degree.

Claim — Let $Q(x)$ be a polynomial satisfying $Q(x) = Q(-x)$. Then $Q(x)$ has no monomials of odd degree.

Proof of the Claim. Note that

$$Q(x) = \frac{Q(x) + Q(-x)}{2} + \frac{Q(x) - Q(-x)}{2}$$

holds. Using $Q(x) = Q(-x)$, it follows that $Q(x) = \frac{Q(x) + Q(-x)}{2}$. Consequently, $Q(x)$ has no monomials of odd degree. $\square$

Remark. The above decomposition of $Q(x)$ is a special case of general phenomena^a.

^aCan you think of a few? Which **general phenomena** is referred to?!

Remark. The above solution is more elegant, and less cumbersome. Moreover, it also highlights the underlying reason, whereas the solution below obscures the conceptual viewpoint.

Solution 2. One can multiply the polynomials to note that

$$1 - x + x^2 - x^3 + \cdots - x^{99} + x^{100}$$

$$\begin{aligned}
&= 1 - x + x^2(1 - x) + x^4(1 - x) + x^6(1 - x) + \cdots + x^{98}(1 - x) + x^{100} \\
&= (1 - x)(1 + x^2 + x^4 + x^6 + \cdots + x^{98}) + x^{100}.
\end{aligned}$$

Using this, we obtain

$$\begin{aligned}
&(1 - x + x^2 - x^3 + \cdots - x^{99} + x^{100})(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\
&= ((1 - x)(1 + x^2 + x^4 + x^6 + \cdots + x^{98}) + x^{100})(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\
&= (1 - x)(1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\
&\quad + x^{100}(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\
&= (1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 - x^{101}) \\
&\quad + x^{100}(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\
&= (1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 - x^{101}) \\
&\quad + x^{100}(x + x^3 + x^5 + \cdots + x^{99}) \\
&\quad + x^{100}(1 + x^2 + x^4 + \cdots + x^{98} + x^{100}) \\
&= (1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 - x^{101}) \\
&\quad + x^{101}(1 + x^2 + x^4 + x^6 + \cdots + x^{98}) \\
&\quad + x^{100}(1 + x^2 + x^4 + \cdots + x^{98} + x^{100}) \\
&= 1 + x^2 + x^4 + x^6 + \cdots + x^{98} + x^{100}(1 + x^2 + x^4 + \cdots + x^{98} + x^{100}),
\end{aligned}$$

which has no monomial of odd degree. ■

Example 1.2. Let n be an even positive integer, and let $p(x)$ be a polynomial of degree n such that $p(k) = p(-k)$ for $k = 1, 2, \dots, n$. Prove that there is a polynomial $q(x)$ such that $p(x) = q(x^2)$.

Walkthrough —

- (a) Note that the polynomial $p(x) - p(-x)$ has degree $< n$ because n is even. Observe that it has at least n roots.

Remark. What would happen if n is not assumed to be even?

Solution 3. Note that the polynomial $p(x) - p(-x)$ has degree at most n , and it vanishes at the integers $-n, \dots, -2, 1, 0, 1, 2, \dots, n$. Thus, it has at least $2n + 1$ roots. It follows that the polynomial is identically zero, that is, the polynomials $p(x), p(-x)$ are equal. This implies that $p(x)$ is equal to $q(x^2)$ for some polynomial $q(x)$. ■

Example 1.3. Determine the remainder when $x + x^9 + x^{25} + x^{49} + x^{81} + x^{121}$ is divided by $x^3 - x$.

Exercise 1.4 (MMO 2015 Grade 9 P6). Do there exist two polynomials with integer coefficients such that each of them has a coefficient with absolute value exceeding 2015, but no coefficient of their product has absolute value exceeding 1?

Summary — Try to come up with enough polynomials $g_1(x), g_2(x), g_3(x), \dots$ and $h_1(x), h_2(x), h_3(x), \dots$ such that each of the products $g_1 g_2 g_3 \dots$ and $h_1 h_2 h_3 \dots$ have at least one coefficient which is **large in absolute value**, and all the coefficients of the product $(g_1 g_2 g_3 \dots)(h_1 h_2 h_3 \dots)$ are at most 1 in absolute value.

Walkthrough —

- (a) Try to come up with a polynomial $P(x)$ whose coefficients are at most 1 in absolute value, and it can be written as a product of enough factors (say $f_1(x), f_2(x), \dots$) such that each of such factor $f_i(x)$ admits a decomposition into the product of two polynomials $g_i(x)$ and $h_i(x)$.
- (b) Can you make sure that the product of the g_i 's, and the product of the h_i 's have to have at least one large coefficient?
- (c) For instance, would taking $g_1(x) = g_2(x) = g_3(x) = \dots = 1 - x$ work for some suitable choice of $h_1(x), h_2(x), \dots$?
- (d) Does taking

$$\begin{aligned} h_1(x) &= 1 + x, \\ h_2(x) &= 1 + x + x^2, \\ h_3(x) &= 1 + x + x^2 + x^3, \end{aligned}$$

etc. work?

- (e) Note that the product of enough g_i 's would have a large coefficient (namely, the coefficient of the second largest power of x). On the other hand, the product of enough h_i 's would have a large coefficient (namely, the coefficient of the power of x).
- (f) What can be said about the absolute value of the coefficients of the product of these two products?

The above seems to work except that having a control on the coefficients of the product $(g_1 g_2 g_3 \dots)(h_1 h_2 h_3 \dots)$ seems hard¹.

Solution 4. Consider the polynomial

$$P(x) = (1 - x)(1 - x^2)(1 - x^4)(1 - x^8) \cdots (1 - x^{2^{2016}}).$$

Since

$$1 + 2 + 2^2 + 2^3 + \dots + 2^{n-1} < 2^n,$$

¹Is it because it fails?

it follows that the coefficients of $P(x)$ are at most 1 in absolute value. Note that

$$P(x) = Q(x)R(x)$$

holds where

$$Q(x) = (1 - x)^{2017},$$

$$R(x) = (1 + x)(1 + x + x^2 + x^3) \cdots (1 + x + x^2 + \cdots + x^{2^{2016}-1}).$$

The coefficient of x^{2016} in $Q(x)$ is equal to 2017, and the coefficient of x in $R(x)$ is equal to 2016. This completes the proof. ■

§2 Factorization and roots

Example 2.1. Let $g(x)$ and $h(x)$ be polynomials with real coefficients such that

$$g(x)(x^2 - 3x + 2) = h(x)(x^2 + 3x + 2)$$

and $f(x) = g(x)h(x) + (x^4 - 5x^2 + 4)$. Prove that $f(x)$ has at least four real roots.

Exercise 2.2 (USAMO 1975 P3, AoPS). [GA17, Problem 151] A polynomial $P(x)$ of degree n satisfies

$$P(k) = \frac{k}{k+1} \quad \text{for } k = 0, 1, 2, \dots, n.$$

Find $P(n+1)$.

Walkthrough —

(a) Consider the polynomial $(x+1)P(x) - x$.

Solution 5. Note that $xP(x+1) - x$ is a polynomial of degree $n+1$, and it vanishes at the $n+1$ integers $0, 1, 2, \dots, n$. It follows that

$$(x+1)P(x) - x = cx(x-1)(x-2) \cdots (x-n)$$

for some nonzero real number c . Substituting $x = -1$ yields

$$-1 = (-1)^{n+1}c(n+1)!,$$

which gives $c = \frac{(-1)^n}{(n+1)!}$. This implies that

$$(n+2)P(n+1) = n+1 + (-1)^n,$$

and consequently,

$$P(n+1) = \frac{n+1 + (-1)^n}{n+2}.$$

■

Exercise 2.3 (Putnam 1999 A2, AoPS). Show that for some fixed positive integer n , we can always express a polynomial with real coefficients which is nowhere negative as a sum of the squares of n polynomials.

Walkthrough —

- (a) Show that the real roots of P have even multiplicity.
- (b) Conclude that P can be expressed as a product of monic quadratic polynomials with real coefficients having nonreal roots, and even powers of linear polynomials with real coefficients.
- (c) Show that a monic quadratic polynomial with real coefficients having nonreal roots is the sum of the squares of two polynomials with real coefficients.

Solution 6. Note that if P is a constant polynomial, then it is clear. Henceforth, let us assume that P is a nonconstant polynomial.

Claim — The polynomial P can be written as the product of polynomials, each of which can be expressed as the sum of the squares of two polynomials with real coefficients.

Proof of the Claim. Since P has real coefficients, it follows that if $\alpha \in \mathbb{C} \setminus \mathbb{R}$ is a root of P , then so is $\bar{\alpha}$. Thus, the nonreal complex roots of P form pairs of complex conjugates. Note that

$$(x - \alpha)(x - \bar{\alpha}) = (x - \operatorname{Re}(\alpha))^2 + \operatorname{Im}(\alpha)^2.$$

Decomposing P over the pairs of nonreal complex conjugate roots, and the real roots, it follows that P can be expressed as the product

$$cf(x) \prod_{a \in A} (x - a)^{m_a},$$

where c denote the leading coefficient of P , $f(x)$ denotes the product of (possibly no) quadratic polynomials of the form $(x - a)^2 + b^2$ with $a \in \mathbb{R}$, $b \in \mathbb{R} \setminus \{0\}$, and A denotes the set of real roots of P , and for an element $a \in A$, the multiplicity of a is denoted by m_a .

Evaluating P at a suitable real number (for instance, at $1 + \sum_{a \in A} a$ (resp. 1) if A is nonempty (resp. empty)), it follows that $c > 0$.

Let a be an element of A . Since A is finite, there exists a real number $\varepsilon > 0$ such that the interval $(a - \varepsilon, a + \varepsilon)$ contains no real roots of P other than a . If m_a were odd, then the sign of $P(x)$ would not remain constant as x ranges over in $(a - \varepsilon, a + \varepsilon) \setminus \{a\}$. Hence, it follows that m_a is even.

Since $c > 0$ and m_a is even for any $a \in A$, the Claim follows. $\square$

Claim — Let $f_1(x), g_1(x), f_2(x), g_2(x)$ be polynomials with real coefficients. Then the following holds.

$$\begin{aligned} & (f_1(x)^2 + g_1(x)^2)(f_2(x)^2 + g_2(x)^2) \\ &= (f_1(x)f_2(x) - g_1(x)g_2(x))^2 + (f_1(x)g_2(x) - f_2(x)g_1(x))^2 \end{aligned}$$

Proof of the Claim. Note that

$$\begin{aligned} & (f_1(x)^2 + g_1(x)^2)(f_2(x)^2 + g_2(x)^2) \\ &= f_1(x)^2 f_2(x)^2 + g_1(x)^2 g_2(x)^2 - 2f_1(x)f_2(x)g_1(x)g_2(x) \\ &\quad + f_1(x)^2 g_2(x)^2 + f_2(x)^2 g_1(x)^2 + 2f_1(x)g_2(x)f_2(x)g_1(x) \\ &= (f_1(x)f_2(x) - g_1(x)g_2(x))^2 + (f_1(x)g_2(x) + f_2(x)g_1(x))^2. \end{aligned}$$

□

Combining the above Claims, and using induction, the result follows. ■

§3 Roots of unity

Exercise 3.1 (USAMO 2014 P1, AoPS). Let a, b, c, d be real numbers such that $b - d \geq 5$ and all zeros x_1, x_2, x_3, x_4 of the polynomial $P(x) = x^4 + ax^3 + bx^2 + cx + d$ are real. Find the smallest value the product

$$(x_1^2 + 1)(x_2^2 + 1)(x_3^2 + 1)(x_4^2 + 1)$$

can take.

See also ??, India Pre-RMO 2012 P17.

Walkthrough —

(a)

Solution 7. Note that

$$\begin{aligned} (x_1^2 + 1)(x_2^2 + 1)(x_3^2 + 1)(x_4^2 + 1) &= P(i)P(-i) \\ &= (1 - b + d)^2 + (a - c)^2 \\ &= (b - d - 1)^2 + (a - c)^2 \\ &\geq 16. \end{aligned}$$

Taking $a = c$ and $b = 5, d = 0$, we obtain

$$(x_1^2 + 1)(x_2^2 + 1)(x_3^2 + 1)(x_4^2 + 1) = 16.$$

Hence, the smallest value the product $(x_1^2 + 1)(x_2^2 + 1)(x_3^2 + 1)(x_4^2 + 1)$ can take is equal to 16. ■

Example 3.2. Let $P(x)$ be a monic polynomial with integer coefficients such that all its zeroes lie on the unit circle. Show that all the zeroes of $P(x)$ are roots of unity, that is, $P(x)$ divides $(x^n - 1)^k$ for some positive integers n, k .

The author has learnt the following argument from [Mainak Ghosh](#), during the [INMO Training Camp 2025](#), held at IISER Bhopal.

Walkthrough —

- (a) Use the fundamental theorem of symmetric polynomials, to prove the following claim.

Claim — Let $f(x)$ be a monic polynomial of degree n with integer coefficients. Let $\alpha_1, \dots, \alpha_n$ denote its roots, counting multiplicities. Then for any integer $k \geq 1$, there is a monic polynomial of degree n with integer coefficients, having $\alpha_1^k, \alpha_2^k, \dots, \alpha_n^k$ as its roots.

- (b) Applying the Claim, for each integer $k \geq 1$, obtain a monic polynomial $P_k(x)$ with integer coefficients, having degree same as that of $P(x)$, whose roots, counted with multiplicities, are the k -th powers of the roots of $P(x)$.
- (c) Note that the polynomials $P_1(x), P_2(x), \dots$ are of the same degree, and the absolute values of the coefficients of any of them are bounded from the above by suitable binomial coefficients, which is smaller than 2^n , where n denotes the degree of $P(x)$. Since these polynomials have integer coefficients, by the pigeonhole principle, it follows that there is a positive integer k_0 such that $P_k(x) = P_{k_0}(x)$ holds for infinitely many positive integers k .
- (d) Enumerate the roots of $P(x)$, and matching the roots of $P_k(x)$ with those of $P_{k_0}(x)$ (for suitable k 's), we obtain a permutation of n letters. By the pigeonhole principle, infinitely many k 's yield the same permutation, which implies that there are positive integers $k \neq \ell$ such that for any root of $P(x)$, its k -th and the ℓ -th powers are equal.

Solution 8. ■

Exercise 3.3 (USAMO 1976 P5, AoPS). If $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ are all polynomials such that

$$P(x^5) + xQ(x^5) + x^2R(x^5) = (x^4 + x^3 + x^2 + x + 1)S(x),$$

prove that $x - 1$ is a factor of $P(x)$.

Summary — The primitive 5-th roots of unity can be used to show that $P(1) = 0$.

Walkthrough —

- (a) Substituting primitive 5-th roots of unity (that is, the 5-th roots of unity other than 1) for x , yields several linear equations in $P(1), Q(1), R(1)$.
- (b) Can $Q(1)$ and $R(1)$ be eliminated to obtain that $P(1) = 0$?

Solution 9. Denote the 5-th root of unity $\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$ by ζ . Substituting $x = \zeta, \zeta^2, \zeta^3$, we obtain

$$\begin{aligned} P(1) + \zeta Q(1) + \zeta^2 R(1) &= 0, \\ P(1) + \zeta^2 Q(1) + \zeta^4 R(1) &= 0, \\ P(1) + \zeta^3 Q(1) + \zeta^6 R(1) &= 0. \end{aligned}$$

Eliminating $R(1)$ from the first two equations yields

$$(1 - \zeta^2)P(1) + \zeta^2(1 - \zeta)Q(1) = 0,$$

and eliminating $R(1)$ from the last two equations yields

$$(1 - \zeta^2)P(1) + \zeta^3(1 - \zeta)Q(1) = 0.$$

Eliminating $Q(1)$ from the above two equations, we obtain $(1 - \zeta)P(1) = 0$, which gives $P(1) = 0$. This shows that $x - 1$ is a factor of $P(x)$. ■

Exercise 3.4 (Leningrad Math Olympiad 1991). A finite sequence $a_1, a_2, \dots, a_n$ is called p -balanced if any sum of the form

$$a_k + a_{k+p} + a_{k+2p} + \dots$$

is the same for any $k = 1, 2, 3, \dots, p$. For instance the sequence

$$a_1 = 1, a_2 = 2, a_3 = 3, a_4 = 4, a_5 = 3, a_6 = 2$$

is a 3-balanced. Prove that if a sequence with 50 members is p -balanced for $p = 3, 5, 7, 11, 13, 17$, then all its members are equal zero.

Summary — Consider the polynomial $\sum_{i=1}^{50} a_i x^i$.

Walkthrough —

- (a) Show that the polynomial vanishes at any p -th root of unity, other than 1, for $p \in \{3, 5, 7, 11, 13, 17\}$.
- (b) How many such roots of unity are there in total?

Solution 10. Let $a_1, a_2, \dots, a_{50}$ be a sequence of complex numbers. Assume that it is p -balanced for $p \in \{3, 5, 7, 11, 13, 17\}$. For an integer $n \geq 1$, denote the root of unity $\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$ by ζ_n . Let $P(x)$ denote the polynomial $\sum_{i=1}^n a_i x^i$.

Let $3 \leq p \leq 17$ be a prime. Since $a_1, a_2, \dots, a_{50}$ is p -balanced, for any $1 \leq \ell < p$, we obtain

$$\begin{aligned} P(\zeta_p^\ell) &= \sum_{k=1}^p (a_k + a_{k+p} + \dots) \zeta_p^{k\ell} \\ &= (a_1 + a_{1+p} + \dots) \sum_{k=1}^p \zeta_p^{k\ell} \\ &= 0, \end{aligned}$$

where the final equality follows since $\zeta_p^\ell \neq 1$. This shows that the polynomial $P(x)$ vanishes at the elements of the set

$$\cup_{p \in \{3, 5, 7, 11, 13, 17\}} \{\zeta_p^\ell \mid 1 \leq \ell < p\},$$

which contains

$$\sum_{p \in \{3, 5, 7, 11, 13, 17\}} (p-1) = 2 + 4 + 6 + 10 + 12 + 16 = 50$$

elements. Moreover, $P(x)$ also vanishes at 0. Note that $P(x)$ is a polynomial of degree 50, and it has at least 51 zeroes. This gives that $P(x) = 0$, and hence, the terms of the sequence $a_1, a_2, \dots, a_{50}$ are all equal to zero. ■

§4 Growth of polynomials

Example 4.1 (India RMO 2015b P3). Find all integers a, b, c such that $a^2 = bc + 4$ and $b^2 = ca + 4$.

Summary — In absolute value, a higher degree polynomial dominates a smaller degree polynomial at arguments which are large enough in absolute value.

Solution 11. Let a, b, c be integers satisfying the given equations.

Let us first consider the case that $a = b$. Note that $a(a - c) = 4$ holds, which shows that $(a, a - c)$ is equal to one of the elements of

$$\{(d, d - 4/d) \mid d \text{ is a divisor of } 4\},$$

and hence (a, b, c) is equal to one of

$$(1, 1, -3), (-1, -1, 3), (2, 2, 0), (-2, -2, 0), (4, 4, 3), (-4, -4, -3).$$

Now, let us consider the case that $a \neq b$. Note that

$$a^3 - b^3 = a(bc + 4) - b(ca + 4) = 4(a - b)$$

holds, which yields

$$a^2 + ab + b^2 = 4.$$

It follows that at least one of a, b is even, and hence, both of them are even. Observe that

$$\left(a + \frac{b}{2}\right)^2 + 3\left(\frac{b}{2}\right)^2 = 4$$

holds, which shows that $(a + b/2, b/2)$ is equal to one of

$$(2, 0), (-2, 0), (1, 1), (1, -1), (-1, 1), (-1, -1),$$

and hence (a, b) is equal to one of

$$(2, 0), (-2, 0), (0, 2), (2, -2), (-2, 2), (0, -2).$$

This implies that (a, b, c) is equal to one of

$$(2, 0, -2), (-2, 0, 2), (0, 2, -2), (2, -2, 0), (-2, 2, 0), (0, -2, 2).$$

Considering the above cases, it follows that (a, b, c) is equal to one of

$$(1, 1, -3), (-1, -1, 3), (2, 2, 0), (-2, -2, 0), (4, 4, 3), (-4, -4, -3), \\ (2, 0, -2), (-2, 0, 2), (0, 2, -2), (2, -2, 0), (-2, 2, 0), (0, -2, 2).$$

Note that any of the above pairs satisfy the given equations. This proves that the above tuples are precisely all the solutions of the given equation over the integers. ■

§5 Size of the roots

Example 5.1. Find all polynomials $P(x)$ (with complex coefficients) satisfying

$$P(x)P(x+2) = P(x^2).$$

Summary — Note that if α is a root of P , then so are α^2 and $(\alpha - 2)^2$. Considering absolute values, show that P cannot have a root other than 1. Conclude that $P(x) = c(x - 1)^n$.

Walkthrough —

(a)

Solution 12. Let $P(x)$ be a polynomial with complex coefficients satisfying the given condition.

Claim — If P has a root $\alpha \neq 1$, then P has a root which has absolute value larger than the absolute value of α .

Proof of the Claim. Note that if $|\alpha| \leq 1$, then

$$\begin{aligned} |(\alpha - 2)^2| &\geq (2 - |\alpha|)^2 \\ &\geq 1 \\ &\geq |\alpha|. \end{aligned}$$

Moreover, if $|(\alpha - 2)^2| = |\alpha|$ holds, then $|\alpha| = 1$ and $\alpha = 2r$ for some real number $r \geq 0$, and hence, $\alpha = 1$. This shows that if $\alpha \neq 1$, and $|\alpha| \leq 1$, then $|(\alpha - 2)^2| > |\alpha|$ holds. Also note that if $|\alpha| > 1$, then $|\alpha^2| > |\alpha|$ holds. Since $\alpha^2, (\alpha - 2)^2$ are also roots of P , the Claim follows. $\square$

If P has a root other than 1, then applying the above Claim to a root of P having the largest absolute value yields a contradiction. Hence, P is a constant polynomial, or the only root of P is equal to 1.

If P is a constant, then P is equal to 0 or 1. If P is a nonconstant polynomial, then P is equal to $c(x - 1)^n$ for some $c \in \mathbb{C} \setminus \{0\}$, and an integer $n \geq 1$. Using the hypothesis, it follows that $c = 1$.

Observing that the polynomials $0, 1, (x - 1)^n$ satisfy the given condition, we conclude that there are all the required polynomials. $\blacksquare$

Example 5.2 (India INMO 2018 P4). Find all polynomials $P(x)$ with real coefficients such that $P(x^2 + x + 1)$ divides $P(x^3 - 1)$.

Walkthrough —

- (a) Show that if α is a root of $P(x)$, then $P(x)$ vanishes at $(\beta_1 - 1)\alpha$ and $(\beta_2 - 1)\alpha$, where β_1, β_2 are the roots of $x^2 + x + 1 = 0$.
- (b) If α is nonzero, then show that one of $(\beta_1 - 1)\alpha$ and $(\beta_2 - 1)\alpha$ is larger than α in absolute value.

Solution 13. Let us establish the following claim.

Claim — Let α denote a nonzero root of $P(x)$ in $\mathbb{C}$. Then for some $\beta \in \mathbb{C}$ satisfying $\beta^2 + \beta + 1 = \alpha$, the element $(\beta - 1)\alpha$ is a root of $P(x)$ and is larger than α in absolute value.

Proof of the Claim. Let β_1, β_2 denote the roots of $x^2 + x + 1 = \alpha$ in $\mathbb{C}$. Since $P(x^2 + x + 1)$ divides the polynomial $P(x^3 - 1)$, it follows that for any $1 \leq i \leq 2$,

$$\beta_i^3 - 1 = (\beta_i - 1)(\beta_i^2 + \beta_i + 1) = (\beta_i - 1)\alpha$$

is a root of $P(x)$. Noting that

$$\begin{aligned} |\beta_1 - 1| + |\beta_2 - 1| &\geq |\beta_1 + \beta_2 - 2| \\ &= |-1 - 2| \\ &> 2, \end{aligned}$$

we obtain that at least one of $\beta_1 - 1, \beta_2 - 1$ has absolute value larger than 1. Consequently, $|(\beta_i - 1)\alpha| > |\alpha|$ holds for some $1 \leq i \leq 2$. This proves the Claim. $\square$

If $P(x)$ has a nonzero root in $\mathbb{C}$, then applying the above Claim to a root of $P(x)$ of largest absolute value, we would obtain a contradiction. Hence, $P(x)$ is equal to cx^n for some $c \in \mathbb{R}$ and an integer $n \geq 0$. Moreover, any polynomial of this form satisfies the given condition. Consequently, these are all the polynomials satisfying the given condition. $\blacksquare$

Remark. Note that proving $|(\beta_1 - 1)(\beta_2 - 1)| > 1$, in order to conclude that at least one of $\beta_1 - 1, \beta_2 - 1$ has absolute value larger than 1, does not seem to work. Moreover, $|(\beta_1 - 1)(\beta_2 - 1)|$ is smaller than $\frac{1}{2}(|\beta_1 - 1| + |\beta_2 - 1|)$. **Unsurprisingly**, a lower bound for the bigger quantity can easily be obtained.

Exercise 5.3 (Problem 4.12, Putnam training problems by Miguel A. Lerma). Does there exist a polynomial $f(x)$ satisfying

$$xf(x - 1) = (x + 1)f(x)?$$

Walkthrough —
(a)

Solution 14. Note that the zero polynomial is the only constant polynomial which satisfies the given condition.

Suppose there exists a nonconstant polynomial $f(x)$ satisfying the given condition.

Claim — Any root of $f(x)$ is an integer.

Proof of the Claim. On the contrary, let us assume that α is root of $f(x)$, and that α is not an integer. Note that for any root β of $f(x)$ in $\mathbb{C} \setminus \mathbb{Z}$, the element $\beta - 1$ is also a root of $f(x)$ lying in $\mathbb{C} \setminus \mathbb{Z}$. It follows that for any integer $k \geq 1$, the element $\alpha - k$ is also a root of $f(x)$. Taking k to be an integer larger than $|\alpha| + \sum_{\gamma \in \mathbb{C}, f(\gamma)=0} |\gamma|$, we obtain

$$|\alpha - k| \geq k - |\alpha| > |\gamma|,$$

and hence $f(x)$ has a root having absolute value larger than that of any of its roots, which is impossible. $\square$

A similar argument can be used to prove that $f(x)$ does not vanish at any negative integer. Moreover, noting that if $f(x)$ vanishes at an integer $n \geq -1$, then $f(x)$ also vanishes at $n+1 \geq -1$, it can be proved using a similar argument that the roots of $f(x)$ are ≤ -2 . This contradicts the assumption that there exists a nonconstant polynomial satisfying the given condition. Consequently, the zero polynomial is the only polynomial that satisfies the given condition. $\blacksquare$

§6 Differentiation and double roots

Lemma 1

Let $P(x)$ be a polynomial with complex coefficients, and α be a complex number. Then α is a root of $P(x)$ having multiplicity at least $r \geq 2$ (i.e., $(x - \alpha)^r$ divides $P(x)$) if and only if it is a root of $P(x), P'(x), \dots, P^{(r)}(x)$, where $P^{(r)}(x)$ denotes the r -fold derivative of $P(x)$.

Solution 15. $\blacksquare$

Example 6.1 (Putnam 1956 B7, IMOSL 1981 Cuba). The polynomials $P(z)$ and $Q(z)$ with complex coefficients have the same set of numbers for their zeroes but possibly different multiplicities. The same is true of the polynomials $P(z) + 1$ and $Q(z) + 1$. **Assume that at least one of $P(z), Q(z)$ is nonconstant.** Prove that $P(z) = Q(z)$.

Walkthrough —

- (a) Assume that $\deg P \geq \deg Q$.
- (b) Denote these two set of roots by S_1, S_2 . Considering multiplicities, show that

$$2 \deg P - |S_1| - |S_2| \leq \deg P' = \deg P - 1,$$

which yields

$$|S_1| + |S_2| > \deg P.$$

- (c) Note that $P - Q$ vanishes at the elements of $S_1 \cup S_2$, which has size larger than the degree of $P - Q$.

Solution 16. On the contrary, let us assume that $P \neq Q$. Without loss of generality, let us assume that $\deg P \geq \deg Q$. Let S_1 (resp. S_2) denote the common set of zeroes of P, Q (resp. $P + 1, Q + 1$). For a polynomial $f(x)$, let us denote its multiset of zeroes by $\mathcal{Z}(f)$.

Note that $\mathcal{Z}(P')$ contains $\mathcal{Z}(P) \setminus S_1$, and $\mathcal{Z}(P')$ also contains $\mathcal{Z}(P + 1) \setminus S_2$. Since $\mathcal{Z}(P)$ and $\mathcal{Z}(P + 1)$ are disjoint, it follows that

$$2 \deg P - |S_1| - |S_2| \leq \deg P' < \deg P - 1,$$

where the final step holds since $\deg P \geq \deg Q$, and one of P, Q is nonconstant. This gives that $|S_1| + |S_2| > \deg P$.

Note that S_1, S_2 are disjoint, and

$$P(z) - Q(z) = (P(z) + 1) - (Q(z) + 1)$$

holds. It follows that $P - Q$ vanishes at $S_1 \cup S_2$, and hence $\deg P \geq |S_1| + |S_2|$. This contradicts the inequality $|S_1| + |S_2| > \deg P$. Consequently, we obtain $P = Q$. ■

§7 Crossing the x -axis/Intermediate value theorem

Exercise 7.1 (China TST 1995 Day 2 P2, AoPS). Alice and Bob play a game with a polynomial of degree at least 4:

$$x^{2n} + \square x^{2n-1} + \square x^{2n-2} + \cdots + \square x + 1.$$

They take turns to fill the empty boxes. If the resulting polynomial has no real root, Alice wins, otherwise, Bob wins. If Alice goes first, who has a winning strategy?

Walkthrough —

- (a) There are more odds than evens among the integers $1, 2, \dots, 2n - 1$.
 (b) Can Bob have a winning strategy using the odds in his favour?

Solution 17. Bob has a winning strategy, as described below.

Bob makes sure that at the end of each of his turns except the last one, the number of even powers of x whose coefficients have been provided by some of them is equal to the number of odd powers of x whose coefficients have been

provided by some of them. This can be done, for instance, if during a turn of Bob, other than the last turn, Bob provides the coefficient of an odd (resp. even) power of x if Alice has provided the coefficient of an even (resp. odd) power of x in the preceeding turn.

Since $n \geq 2$, it follows that Bob gets at least one turn. At the beginning of the final turn of Bob, there are two powers of x whose coefficients are to be determined, denote them by x^i, x^j , their coefficients by c_i, c_j respectively. Let $Q(x)$ denote the polynomial, obtained by the erasing the terms corresponding to x^i, x^j from the polynomial that Bob had at the beginning of his final turn. Note that

$$P(x) = Q(x) + c_i x^i + c_j x^j.$$

Note that at least of i, j is odd. Interchanging i, j if necessary, let us assume that i is odd. We describe the strategy that Bob follows in the two cases below.

Let us consider the case that j is even. Bob determines c_j in such a way that for any choice of c_i , the completed polynomial $P(x)$ is guaranteed to have at least one real root. This can be done, for instance, by taking c_j satisfying

$$Q(1) + Q(-1) + 2c_j = 0.$$

For any choice of c_i , the above choice of c_j shows that $P(1) + P(-1) = 0$, which implies that $P(x)$ has a root in the interval $[-1, 1]$.

Let us consider the case that j is odd. Bob determines c_j in such a way that for any choice of c_i by Alice in the next turn, the completed polynomial $P(x)$ is guaranteed to have at least one real root. This can be done, for instance, by taking c_j satisfying

$$Q(2) + c_j 2^j + 2^i Q(-1) - c_j 2^i = 0.$$

Since $i \neq j$, the above holds for some $c_j \in \mathbb{R}$. For any choice of c_i , note that

$$P(2) + 2^i P(-1) = 0$$

holds, which implies that $P(x)$ has a root in $[-1, 2]$. ■

§8 Lagrange interpolation

Example 8.1. If a polynomial of degree n takes rationals to rationals on $n + 1$ points, then show that it is a rational polynomial.

Walkthrough —

- (a) Apply the Lagrange interpolation formula.

Solution 18. ■

Exercise 8.2 (USAMO 2002 P3, AoPS). Prove that any monic polynomial (a polynomial with leading coefficient 1) of degree n with real coefficients is the average of two monic polynomials of degree n with n real roots.

Walkthrough —

- (a) Let $F(x)$ be a monic polynomial of degree n with real coefficients. We would like to write

$$2F(x) = P(x) + Q(x),$$

where $P(x), Q(x)$ are polynomials with certain properties.

- (b) Let us take $P(x)$ to be a polynomial which changes sign very often, so that it is likely to have n real roots. To do so, choose n real numbers satisfying

$$x_1 < x_2 < \cdots < x_n,$$

and let $y_1, \dots, y_n$ be real numbers (to be specified later). Apply the Lagrange interpolation formula to obtain a monic polynomial $P(x)$ satisfying $P(x_i) = y_i$ for all i .

- (c) Define the polynomial $Q(x)$ using

$$2F(x) = P(x) + Q(x).$$

Note that $Q(x)$ is a monic polynomial with real coefficients.

- (d) Can one impose suitable conditions on $y_1, \dots, y_n$ such that $Q(x)$ changes sign often?

Solution 19. Let $F(x)$ be a monic polynomial of degree n with real coefficients. Let $x_1 < \cdots < x_n$ be real numbers. Note that there exist real numbers $y_1, \dots, y_n$ satisfying

$$(-1)^i y_i > 0, \quad (-1)^{i-1} (2F(x_i) - y_i) > 0$$

for any $1 \leq i \leq n$. Indeed, y_i 's can be taken satisfying $(-1)^i y_i > |F(x_i)|$. Let $P(x)$ be a monic polynomial of degree n with real coefficients such that $P(x_i) = y_i$ for all $1 \leq i \leq n$. Let $Q(x)$ denote the polynomial such that $F(x)$ is the average of $P(x)$ and $Q(x)$. Since $F(x), P(x)$ are monic, it follows that so is $Q(x)$. For any $1 \leq i \leq n$, note that $Q(x_i) = 2F(x_i) - P(x_i) = 2F(x_i) - y_i$ holds, and hence $Q(x_i)$ has sign as that of $(-1)^i$. It follows that each of the polynomials $P(x), Q(x)$ has at least one root in each of the intervals

$$(x_1, x_2), (x_2, x_3), \dots, (x_{n-1}, x_n).$$

Since these polynomials are of degree n with real coefficients and each of them has at least $n - 1$ real roots, all of their roots are real. ■

Exercise 8.3 (Putnam 1968 A6, AoPS). Find all polynomials whose coefficients are all ± 1 and whose roots are all real.

Walkthrough —

- (a) Consider the average of the squares of the roots, and show that it is small (and consequently, smaller than their geometric mean) if the polynomial has degree ≥ 4 .
- (b) Repeat the argument for degree three polynomials.
- (c) Finding the degree one and degree two polynomials is easy.

Solution 20. Let $P(x)$ be a polynomial of degree n having real roots. Assume that its coefficients are equal to ± 1 . Denote its roots by $\alpha_1, \dots, \alpha_n$, counting multiplicities. Noting

$$\alpha_1^2 + \dots + \alpha_n^2 = (\alpha_1 + \dots + \alpha_n)^2 - 2 \sum_{1 \leq i < j \leq n} \alpha_i \alpha_j,$$

it follows that

$$\alpha_1^2 + \dots + \alpha_n^2 = 3.$$

Applying the AM-GM inequality, we obtain

$$\alpha_1^2 + \dots + \alpha_n^2 \geq n,$$

which implies that $n \leq 3$.

If $P(x)$ is a monic linear polynomial, then it is equal to one of $x - 1, x + 1$.

If $P(x)$ is a monic quadratic polynomial, then considering discriminants, it follows that its constant term is equal to 1, and hence $P(x)$ is equal to one of $x^2 + x - 1, x^2 - x - 1$.

Let us consider the case that $P(x)$ is a monic cubic polynomial, and the coefficient of x^2 in $P(x)$ is equal to 1. Note that $x^3 + x^2 + x - 1$ vanishes at $x = 0$. For any real number $\alpha \leq 0$,

$$\alpha^3 + \alpha^2 + \alpha - 1 = \alpha(\alpha + 1)^2 - 1 - \alpha^2 \leq -1$$

holds, which implies that any real root of $x^3 + x^2 + x - 1$ is positive. Since $x^3 + x^2 + x - 1$ is of odd degree, it has a real root, and since $x \mapsto x^3 + x^2 + x - 1$ is an increasing function on the set of positive reals, it follows that $x^3 + x^2 + x - 1$ has only one real root. This gives $P(x) \neq x^3 + x^2 + x - 1$.

Let us consider the case that $P(x) = x^3 + x^2 - x + 1$ Using

$$P(x) = (x^2 - x) + 1 + x^3,$$

$$P(x) = x^3 + x^2 + (1 - x),$$

$$P(x) = x^3 + (x^2 - x) + 1,$$

it follows that $P(x)$ has no root in $[-1, \infty)$. Note that for $-1 \geq a \geq b$, we have

$$P(a) - P(b) = (a - b)(a^2 + ab + b^2 + a + b - 1)$$

$$\begin{aligned}
&= (a-b)((a^2+a) + (b^2+b) + (ab-1)) \\
&\geq 0.
\end{aligned}$$

This shows that x^3+x^2-x+1 has no real root, and hence $P(x) \neq x^3+x^2-x+1$.

Note that not all the roots of

$$x^3+x^2+x+1 = (x+1)(x^2+1)$$

are real. Consequently, if $P(x)$ is a monic cubic polynomial and the coefficient of x^2 in $P(x)$ is equal to 1, then $P(x)$ is equal to x^3+x^2-x-1 .

If $P(x)$ is a monic cubic polynomial and the coefficient of x^2 in $P(x)$ is -1 , then $-P(-x)$ is equal to x^3+x^2-x-1 , and hence $P(x)$ is equal to x^3-x^2-x+1 .

This shows that $P(x)$ is equal to one of

$$x-1, x+1, x^2+x-1, x^2-x-1, x^3+x^2-x-1, x^3-x^2-x+1.$$

Note that the discriminants of the quadratic polynomial x^2+x-1, x^2-x-1 are nonnegative, and also note that

$$\begin{aligned}
x^3+x^2-x-1 &= (x-1)(x+1)^2, \\
x^3-x^2-x+1 &= (x-1)^2(x+1).
\end{aligned}$$

Consequently, the roots of the polynomials

$$x-1, x+1, x^2+x-1, x^2-x-1, x^3+x^2-x-1, x^3-x^2-x+1$$

are all real. Hence, the required polynomials are

$$\begin{aligned}
&x-1, x+1, x^2+x-1, x^2-x-1, x^3+x^2-x-1, x^3-x^2-x+1, \\
&-(x-1), -(x+1), -(x^2+x-1), -(x^2-x-1), -(x^3+x^2-x-1), -(x^3-x^2-x+1).
\end{aligned}$$

■

§9 Integer divisibility

Example 9.1. If P is a polynomial with integer coefficients and a, b are integers, then $P(a) - P(b)$ is a multiple of $a - b$.

Walkthrough —

- (a) Show that it suffices to prove it for monomials.

Solution 21.

■

Exercise 9.2 (USAMO 1974 P1, AoPS). Let a, b , and c denote three distinct integers, and let P denote a polynomial having all integral coefficients. Show that it is impossible that $P(a) = b$, $P(b) = c$, and $P(c) = a$.

Walkthrough —

- (a) Show that each of the integers $a - b, b - c, c - a$ is a multiple of the remaining two integers.
- (b) Prove that this implies that a, b, c are equal.

Solution 22. Note that

$$a - b \mid P(a) - P(b) = b - c \mid P(b) - P(c) \mid c - a \mid P(c) - P(a) = a - b.$$

Consequently, the integers $a - b, b - c, c - a$ are of the same absolute value. Denote their absolute value by k . Note that their sum is zero. However, the sum is equal to mk , for some $m \in \{\pm 1, \pm 3\}$. Hence, k is equal to zero. This yields that $a = b = c$. ■

Here is a more general result.

Example 9.3. Let $P(x)$ be a polynomial with integer coefficients, and let n be an odd positive integer. Suppose that $x_1, x_2, \dots, x_n$ is a sequence of integers such that $x_2 = P(x_1), x_3 = P(x_2), \dots, x_n = P(x_{n-1})$, and $x_1 = P(x_n)$. Prove that all the x_i 's are equal.

Walkthrough — Show that

$$a_1 - a_2 \mid a_2 - a_3 \mid a_3 - a_4 \mid \dots \mid a_n - a_1 \mid a_1 - a_2.$$

Note that sum of these differences is an odd multiple of their absolute value.

Lemma 2

Let P be a polynomial with integer coefficients. Suppose a is an integer and k is a positive integer such that $P^k(a) = a$, where P^k denotes the k -fold composite map from $\mathbb{Z} \rightarrow \mathbb{Z}$. Show that $P^2(a) = a$.

Proof. Let ℓ denote the smallest positive integer such that $P^\ell(a) = a$. If $\ell = 1$ or $\ell = 2$, then we are done. Henceforth, we assume that $\ell \geq 3$.

Note that

$$P(a) - a \mid P^2(a) - P(a) \mid \dots \mid P^\ell(a) - P^{\ell-1}(a) = a - P^{\ell-1}(a) \mid P(a) - a.$$

Since $a - P^{\ell-1}(a)$ is nonzero, it follows that the above differences are nonzero. Consequently, for any $1 \leq i \leq \ell$,

$$P^{i+1}(a) - P^i(a) = \pm(P^i(a) - P^{i-1}(a)).$$

If $P^{i+1}(a) = P^{i-1}(a)$ holds for some $1 \leq i \leq \ell$, then applying $P^{\ell-i+1}$ to both sides, we obtain $P^2(a) = a$, which contradicts the assumption that $\ell \geq 3$. It follows that for any $1 \leq i \leq \ell$,

$$P^{i+1}(a) - P^i(a) = P(a) - a$$

holds, which implies that

$$\sum_{i=0}^{\ell-1} (P^{i+1}(a) - P^i(a)) = \ell(P(a) - a).$$

This gives $P(a) = a$, which contradicts the assumption that $\ell \geq 3$. This completes the proof. $\square$

Exercise 9.4 (IMO 2006 P5, AoPS, by Dan Schwarz, Romania). Let $P(x)$ be a polynomial of degree $n > 1$ with integer coefficients, and let k be a positive integer. Consider the polynomial $Q(x) = P(P(\dots P(P(x)) \dots))$, where P occurs k times. Prove that there are at most n integers t such that $Q(t) = t$.

Walkthrough —

(a) Does the above lemma help?

Solution 23. By the above lemma, it reduces to considering the case $Q(x) = P^2(x)$.

Suppose Q has more than n fixed points. Since P is not linear, it follows that P cannot have n fixed points, and hence not all the fixed points of Q are fixed points of P . Let b be a non-fixed point of P , and $Q(b) = b$. Suppose a be a fixed point of Q , other than b .

Let us first consider the case that $P(a) \neq a$. Note that

$$P(b) - a \mid P(a) - b \mid P(b) - a$$

holds, and

$$a - b \mid P(a) - P(b) \mid a - b$$

holds too. This yields that

$$|P(a) - b| = |P(b) - a|, \quad |P(a) - P(b)| = |a - b|.$$

If

$$P(a) - b = a - P(b), \text{ and } P(a) - P(b) = a - b$$

hold, then b would be a fixed point of P . It follows that at least one of

$$P(a) - b = -(a - P(b)), P(a) - P(b) = -(a - b)$$

holds. Consequently, we obtain

$$P(a) + a = P(b) + b.$$

Next, let us consider the case that $P(a) = a$. Note that

$$P(b) - a \mid b - a \mid P(b) - a.$$

Since b is not a fixed point for P , it follows that

$$P(b) - a = a - b,$$

which yields

$$P(a) + a = P(b) + b.$$

This proves that all the roots of $Q(x) = x$ are the roots of $P(x) + x = P(b) + b$. Since $P(x)$ has degree $n > 1$, it follows that the polynomial $P(x) + x - P(b) - b$ is of degree n , and it has more than n roots, which is impossible.

Hence, there are at most n integers t such that $Q(t) = t$ holds. ■

References

- [GA17] RĂZVAN GELCA and TITU ANDREESCU. *Putnam and beyond*. Second. Springer, Cham, 2017, pp. xviii+850. ISBN: 978-3-319-58986-2; 978-3-319-58988-6. DOI: [10.1007/978-3-319-58988-6](https://doi.org/10.1007/978-3-319-58988-6). URL: <https://doi.org/10.1007/978-3-319-58988-6> (cited p. 5)